Travel Web App

Project Management Plan

VERSION 1.0.0

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TEAM 074

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Introduction and Purpose

Student exchange programs are opportunities available at many universities in which students get the chance to study abroad. Students that take part in these programs do not only receive an education in another country but often travel within that country to be immersed in the foreign lifestyle, language, and culture. Consequently, Monash University has sponsored the creation of a travel web application that will assist university students in planning their domestic travels around a country. Although currently international borders are closed due to the current COVID-19 pandemic, the federal government is taking steps to ease these restrictions, and it is believed this application will contribute to the recovery of the economies.

The purpose of this document is to outline all the necessary information about the project for those internal members who will be working on the project. This document outlines the necessary information on the new travel web application. Additionally, it clearly outlines who is involved in the project and how the project will be executed, monitored, and controlled to produce the desired travel application for Monash University.

Project Information

Background and Intended Use

Currently, it is assumed that there is no application that allows university exchange students to easily plan travels within a country. Additionally, organising travels can be complicated and confusing for many students who have never been overseas. Consequently, there is a strong need for this travel app. While it is being created on behalf of Monash University, it is assumed that this travel app will be able to be accessed by all university students who decide to go on a student exchange program offered by any university. Hence, while the application is being developed for Lauren from Monash University (the client), this application will be used by university students who agree to this cultural immersion opportunity.

The travel app is supposed to be used on devices – including smartphones, tablets and computers – of which may have different operating systems. It is intended to allow students who want to plan domestic trips in any country to choose routes and add connection routes (if desired); view, cancel and save trips to an account (if desired); as well view any past trips that they have gone on.

Scope

The user-friendly travel web application is required to allow users to search for their exchange country and select a date for which they want to plan a trip. It is expected that both the country and date inputted is valid. A map of the selected country with all the airports should be shown. After selecting an airport that they want to depart from, all the possible domestic flights leaving that airport should be shown. The user should be able to select a flight and get a general overview of the flight details including starting location, end location, the airline and the number of stops. However, it is out of scope to check whether there are any seats available or to compare different airlines and airline prices. Additionally, it is out of scope to provide more specific detail such as departure times and arrival times. These are excluded as these finer details will be managed by an external booking agency and booking system. After adding a route, the user should be able to add as many additional connection flights as they would like and then confirm the trip. Once confirmed, a summary of the trip is expected to be shown and the user should have the option to save the trip to their account or to plan a new trip. It is out of scope to allow the user to undo confirming the trip since it is in scope to make sure that the user is certain that they want to finalise their trip via an alert to confirm their request. If the user does not like the trip made, they do not need to save it to their account and they can plan a new trip. If they desire to save the planned trip to their account and the user is logged in, they will be taken directly to a page which displays a summary of all the trips they have saved. The user can select any saved trip to get more details. Additionally, they can cancel a saved trip if desired. If the user is not logged in, before they can save the trip they created in the homepage, they will be requested to log in. In the log in page, a valid email and password is required. Additionally, the user does have the option to not log in, but they will then be taken back to the homepage. The user also has the option to create an account if they do not have one already. If they choose to do so, all they need to provide is their full name, email and password. It is in scope to make sure that the password entered is appropriate. However, it is out of scope to validate the email provided and no email is required to be sent to them to confirm their account creation as the client wants the user to be able to easily and quickly create an account. Consequently, it is also out of scope to validate their identity. Using the navigation bar, the user is also able to view past trips that they have gone on. Similar to the saved trips page, this page displays a summary of all the past trips and the user can, similarly, click on a past trip to get more detail about it. When logged in, the user is also able to use the navigation bar to access their account details. Consequently, the user can change any account details, including their name, email and/or password. Additionally, they have the option to log out of their account. There is no limitation on the number of trips the user can save to their account. Furthermore, it is expected that the students should be able to use the web application on different devices that may have different operating systems. However, it is out of scope for the user to log into the same account on different devices as it is assumed that this would require an extensive amount of cost and time. Since all the information will be stored on each device in local storage, it is also out of scope that the data needs to be encrypted. It is expected that, no matter what page the user is on, they should be able to access a help button that allow them to seek further assistance if required.

Deliverables/Due Dates

The final version of the Travel web application's code is due on Monday, 2 November 2020 at 8:00 AM. Additionally, a 15-minute presentation will be held at 9:00 AM on Wednesday, 4 November 2020. The slides for the presentation has to be submitted before 9:00 AM on Wednesday, 4 November 2020. An overall timeline of all the items due can be seen in Table 1.

What is Due	Commencement Date	Due Date	Dependency
Client Requirements Interview	09/09/2020	09/09/2020	N/A
Requirements Document	09/09/2020	16/09/2020	Client Requirements Interview
Design Document	16/09/2020	23/09/2020	Requirements Document
Project Management Plan	16/09/2020	23/09/2020	Requirements Document
Interface Prototype	23/09/2020	07/10/2020	Design Document
Final App Prototype	07/10/2020	21/10/2020	Interface Prototype
Test Plan Document	21/10/2020	28/10/2020	Final App Prototype
Client Presentation	28/10/2020	04/11/2020	Requirements Document, Final App Prototype, Test Plan Document

Table 1: Due Dates

Personnel/HR Management

Table 2 contains all the necessary information about the team members who will be working on this project.

Name	Contact Details	Responsibilities
Grishma, Khadka	email: grishma.khadka1@monash.edu	Manager - Check the work done by members to ensure the team is on track
Jason, Abi Chebli	email: jabi0003@student.monash.edu	Team leader - Facilitate meetings - Delegate tasks - Work on tasks assigned - Prepare for meetings according to agenda - Participate in discussions actively during meetings
Chansophorn Panha, In	email: cinn0001@student.monash.edu	Minutes taker - Record minutes for meetings. - Work on tasks assigned by the team leader - Prepare for meetings according to agenda - Participate in discussions actively during meetings
Shakthi, Prashanth Champaka	email: spra0027@student.monash.edu	- Work on tasks assigned by the team leader - Prepare for meetings according to agenda - Participate in discussions actively during meetings
Calista Rou Mae, Saw	email: csaw0002@student.monash.edu	- Work on tasks assigned by the team leader - Prepare for meetings according to agenda - Participate in discussions actively during meetings

Table 2: Team Members Details and Responsibilities

Decision on Processes

In this process, the team has to design and implement code to produce a web app, to help the process to be carried out smoothly, a few decisions are made on this.

The three main tools that are to be used in this project are Google Docs, Git and Trello.

Google Docs

This tool is used by the team mainly to store documents such as requirements and design documents. The team is also able to edit the document together to increase efficiency. In addition, Google Docs is used during meetings to write and store meeting minutes. The minutes should be reviewed every time a member is unclear of what is discussed during meetings. This is so that all members can ensure they are clear of what is their task for that period of time and to make sure all members are on the same page. Lastly, Google Docs is used to share all documents related to the project with the manager so that the manager can keep track of the progress made and make sure the team is on track.

Git

The code written by team members is stored in this tool. With this tool, team members can work remotely in their own time and update the code written to Git so that all members can see the code written by other members. Team members should update all code written as often as possible, usually after each session of coding, so that all members can be up to date with code which eases the process of coding for the other members.

Trello

The tasks are tracked by using this software. This software has a kanban style layout which lets all team members have a clear look of all the tasks that are done or yet to be done. Furthermore, this style of task tracking that is shared with all members provides information of progress of each member so that all members are aware of the progress of each member and whoever needs help with completing tasks. Members are expected to utilise Trello every time a task is completed and also when there is a new task to update the other members on their progress.

Communications Management

The following methods of communication are to be used by all team members in this project:

Meetings

The team is to have weekly meetings on Zoom for the purposes of checking the progress and work of each team member with their assigned tasks. Members must participate in every meeting and join within five minutes, and if tardy or absent, must let other members know via written communication prior to the meeting. During each meeting, each member provides a run-through of their work to everyone so that everyone has a shared understanding. Any questions or concerns are to be addressed and resolved in the meeting by taking suggestions and contributions from every team member.

Secondly, any outstanding work is to be allocated to everyone, and completed by a set date. Before doing so, the team leader should go through the requirements of the work to be completed to ensure that everyone has an understanding of what is expected and required, as well as set out the format of the document, if there is any written work to be done. Moreover, the level of difficulty as well as time required for the tasks are estimated through a discussion with members, after which members are free to put forward a choice for the tasks they would like to complete, in order to ensure fairness and equal distribution of work.

Lastly, to conclude each meeting, the team will decide a suitable time to meet, in case any member is unable to meet at the usual time.

Written Communication

The team is to use the Messenger group chat to communicate any issues or questions that have arisen before or after the meetings. Possible topics of discussion may include rescheduling of meetings, questions about the tasks assigned to individual members, to name but a few. It must be noted that discussion topics via written communication should be trivial enough such that it is not necessary for a meeting. Moreover, there is no strict frequency on written communication, however, it is encouraged that members are kept up-to-date of each other’s progress. Each member is expected to respond to messages within half a day, however if there is something due within the upcoming few days, members are expected to respond within 3 hours so that efficiency is not affected. If the response time is exceeded, members will first receive a warning. If this continues to happen, the issue will be brought up with the manager.

Written Tasks

Written tasks involve producing a report or document. Therefore, team members are to use Google Docs within the team Google Drive to work on these tasks, which is also shared to the manager to be assessed. As specified in the meetings section, the format of the document is set out before anyone begins their allocated work to ensure uniformity of the document. The use of Google Docs is to allow each member to see the work of other members as well as their progress on the tasks allocated to them. Any suggestions can then be raised before the next meeting to increase efficiency.

Members must complete the allocated parts before the due date or the next meeting, whichever comes first. Half a day to one day lateness has been allowed to complete the task. If a member is struggling to complete their assigned tasks, they must let other members know as soon as possible via the Messenger group chat and the work can then be redistributed if the deadline is approaching to avoid late submission.

Trello

Team members must make use of Trello, at least weekly, in order to record tasks that are to be completed, tasks in progress and tasks completed. Each task can be moved to the different lists so it is easy to determine the outstanding tasks, and keep the Trello boards organised. This is especially useful not only for team members but also the manager to track the progress of the weekly tasks that have been assigned as well as the entire project.

Risk Management

The risk management plan includes a set of rules, guidelines and protocols in order to predict the risks associated and estimate its consequences and impacts. These also involve mitigation strategies to avoid any further problems.

Table 4 outlines the risk's involved for this project whereas Table 3 defines the different level's that are used in Table 4.

LEVEL	**PROBABILITY**
Almost Certain	Likely to occur every day during the project.
Likely	Likely to occur weekly during the project.
Possible	Might occur once every fortnight.
Unlikely	Likely to occur less than once in a month.
Rare	Unlikely to occur not even once throughout the project.

Table 3: Probability Definitions

Risk	**Consequence**	**Probability**	**Severity**	**Mitigation**
Lack of people with required technical skills.	The work will pile up for the people who possess the required technical skills.	Possible	Moderate	Train and learn the required skills before the start of the project.
Unclear Requirements	The developers will be confused and be in a dilemma about creating the desired application. Consequently, they might create an undesired application.	Unlikely	Major	During the requirement gathering phase, get all the requirements from the client. Not to make any assumptions in the requirement.
Improper allocation of the work among the team members	As a result, few members might get excessive work and might have more workload. Which might affect the performance of that member and resulting in the delay of the project.	Rare	Minor	Proper and equal share of work amongst all the team members considering their strengths and weaknesses.
Miscommunication within the team members	This will cause a deviation from the required application requested by the client. The client may be unsatisfied and might ask for it to be redeveloped.	Possible	Major	Proper communication within team members, organize regular meetings to track everyone’s progress in their tasks.
Major requirements changes post the requirements phase	Any requirements changes might result in not getting the expected application by the client.	Rare	Major	Remain updated about the project requirements. Keeping objectives in every phase to stay organized.

Table 4: Risk Management

Quality Assurance (QA)

The purpose of this plan is to specify how Software Quality Assurance (SQA) will be performed for the project. This plan describes the SQA activities to be performed and designates a set of standardized techniques for performing those activities.
The following describes the functional groups/people that influence and control software quality.

(1) Project Manager

The Project Manager is responsible for determining that staff assigned to the project understand and comply with the Quality Check (QC) procedures that apply to their activities. The Project Manager's QA-related responsibilities are to:

- Review and approve project controlling documents, including requirements document, design document, wireframes and storyboard.
- Communicate project scope requirements to project team members.
- Communicate with the client for feedback on service satisfaction.
- Ensure that project deliverables and activities are in accordance with project controlling documents.
- Respond to corrective action requests and assure that deficiencies are corrected in a timely manner.

(2) Task Manager

A Task Manager is a professional responsible for a particular project task. The Task Manager's responsibilities are to:

- Prepare project technical methods for the project tasks.
- Oversee data collection and analysis and determine that appropriate procedures are employed.
- Prepare reports or report sections related to the project tasks.
- Ensure that all task deliverables are in accordance with project controlling documents.
- Correct deficiencies identified for the project tasks.
- Keep communicating with Project Manager for project requirements and changes in requirements.

QUALITY RELATED DOCUMENTATION

Meeting Minutes

- Meeting minute documentation captures notable decisions made, issues to be tracked, and action items to be completed.
- This document is used to keep track of the work being done and set new objectives for the future meetings.

Kanban Boards Via Trello

The Kanban board includes three features-

- To do
- Doing
- Done

These can be used to meticulously plan the project, without missing out on any part of the project.

This can be used by the project manager to have an idea of the progress made by every team member on a regular basis.